Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Saman Zonouz	Email	szonouz6@gatech.edu
Institute Name	Georgia Institute of Technology			
Address	Atlanta, Georgia, USA			

2. Technology Overview

Title of Proposed Technology	Secure and Resilient Manufacturing Platforms for Industry 4.0: IP Protection, Criticality-Aware Security Assessment, and Impact-Driven Recovery				
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (v), Quantum ( )				
Industrial Technology Classification 1)	Technology Field 1 (Large Category)			Technology Field 2 (Large Category)	
	Cybersecurity			AI	
	Medium Category		Small Category	Medium Category	Small Category
	Manufacturing		Industrial IoT	Cyberphysical system security	
Concept of Proposed Technology	In Industry 4.0 environments, cyber attacks can cause long production downtime with significant financial impact, degrade product quality, damage manufacturing equipment, and in severe cases pose risks to human safety. The proposed project will develop technologies and methodologies to secure modern manufacturing systems against emerging cyber-physical threats, building on existing research products at TRL 3–4 that provide a strong foundation for advancing these solutions to higher technology readiness levels through system integration, validation, and pilot-scale deployment. Key focus areas include:				

(max 10 pages)

	AI-based Side-Channel Attack Modeling and Defenses for IP Protection. Industry 4.0 manufacturing systems rely on side-channel signals such as acoustic, electromagnetic, optical, and power emissions for smart monitoring, state estimation, and closed-loop control. While these signals are necessary for efficient and autonomous operation, they also create an attack surface that can be exploited to leak sensitive manufacturing intellectual property. My research has shown that optical side-channel information from additive-manufacturing devices can be analyzed using machine learning to reconstruct proprietary data such as CAD designs and G-code. Building on this, we will develop machine learning models to identify and quantify side-channel leakage and develop mitigations while preserving the operational role of these signals. The outcome will be a real-time monitoring and defense system that detects potential side-channel surveillance and dynamically activates mitigation strategies to prevent IP leakage without disrupting manufacturing performance. **Vulnerability Assessment and Patching using Agentic AI.** Modern manufacturing systems consist of highly interconnected cyber-physical components, including controllers, sensors, machines, and industrial networks, which together create a large and complex attack surface. This objective focuses on systematically identifying devices and workflows that are both highly vulnerable and critical to manufacturing operations, with respect to potential physical outcomes such as defective products, equipment damage, and safety hazards, using models of the manufacturing process. By combining firmware reverse engineering, ICS vulnerability databases, and threat modeling, the toolchain will identify weaknesses in PLC logic and configuration. The outcome will be a criticality map of the production line that highlights the most critical and vulnerable PLCs, sensors, and processes whose compromise could be catastrophic for the manufacturing systems. My research group has developed multiple vulnerability-assessment and patching solutions for different ICSs, with more recent methods leveraging agentic AI for vulnerability aggregation and analysis as well as developing domain-specific defenses to patch the systems . **Impact Quantification and AI-Based Automated-Recovery Mechanisms.** In real-world manufacturing environments, attack detection, diagnosis, and patching can take significant time, during which the manufacturing system may continue operating and transition into unsafe or unstable states. In this research objective, we will focus on developing an AI response system that can generate automated recovery actions to minimize damage due to a cyber attack. Based on the impact and vulnerability assessment in the previous research objective, our AI response system will generate prioritized responses that address the most dangerous attacks first. Recovery techniques will include software rollback (e.g., restoring PLC code or settings to a last known safe state if tampering is detected) and control-theoretic responses that will try mission-restoration first and then drive the system to a safe state if mission completion is not possible.
Rationale for Joint R&D Project	Industry 4.0 manufacturing systems increasingly rely on tightly integrated cyber-physical components, advanced sensing, and networked control to achieve efficiency, flexibility, and automation. While these capabilities enable smart monitoring and control, they also introduce new attack surfaces that expose manufacturing intellectual property, critical workflows, and physical processes to cyber threats. Recent

(max 10 pages)

	advances in AI-enabled side-channel attacks, industrial control system malware, and stealthy process manipulation demonstrate that traditional IT-centric security approaches are insufficient for protecting modern manufacturing environments. A joint R&D effort is necessary to address these challenges through a holistic, cyber-physical security approach that integrates attack modeling, criticality-aware assessment, and impact-driven recovery. Collaboration between a leading research institution and Korean manufacturing SMEs enables the development of solutions that are both scientifically rigorous and operationally practical. Georgia Tech contributes expertise in cyber-physical system security, side-channel analysis, and control-aware defenses for manufacturing systems, while Korean SME partners provide realistic manufacturing platforms, operational constraints, and deployment environments. This partnership ensures that the resulting technologies are deployable in real Industry 4.0 settings and aligned with the needs of manufacturing startups and SMEs.
Key Words	Provide key terms that describe the proposed technology and its main features (min 3, max 5) 1. Industry 4.0 Manufacturing Security 2. AI-Based CPS Security 3. Side-Channel Attack Detection and IP Protection 4. Criticality-Aware Vulnerability Assessment 5. Impact-Driven Automated Recovery

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		Start of the Project	Level ( )	End of the Project	Level ( )
R&D Objectives		The overarching objective of this project is to enhance the security and resilience of Industry 4.0 manufacturing platforms by using AI to integrate intellectual-property protection, criticality-aware security assessment, and impact-driven automated recovery. Specifically, the project aims to (i) protect sensitive manufacturing data from AI-enabled side-channel attacks while preserving the operational role of side channels in monitoring and control, (ii) systematically identify and secure critical devices and workflows whose compromise could lead to severe physical and operational consequences, and (iii) enable rapid, automated recovery from cyber incidents by quantifying cyber-physical impact and triggering appropriate mitigation actions. Together, these objectives establish a comprehensive cyber-physical security framework for modern manufacturing systems.			

(max 10 pages)

R&D Target Specifications		The criteria to measure success in the proposed research include: **IP Protection Effectiveness:** Demonstrated reduction in the ability of adversaries to reconstruct CAD designs, toolpaths, or machine instructions from side-channel signals under defended configurations. **Coverage and Accuracy of Criticality Analysis:** Ability to correctly identify and prioritize critical devices, workflows, and network links based on vulnerability and potential physical impact. **Impact Assessment Fidelity:** Accurate prediction and detection of cyber-physical impacts, including quality degradation, process disruption, and safety-relevant deviations. **Recovery Performance:** Reduction in time to safe state and time to resume normal operation following simulated cyber attacks, with minimal human intervention. **Deployability:** Successful integration and evaluation of the proposed tools on commercial manufacturing platforms under realistic operational constraints.
R&D Plan		**Milestone 1 – Year 1: Threat Modeling and Baseline Prototypes** Develop baseline AI-enabled side-channel attack models Perform preliminary vulnerability assessment of devices and networks and define threat and impact models for critical workflows. Develop a baseline AI-recovery system. **Milestone 2 – Year 2: Criticality-Aware Security and Impact Analysis** Harden the remote-monitoring systems to minimize side-channel leakage Validate the automated vulnerability assessment in simulation and small testbeds. Validate AI-recovery system in simulation and small testbeds **Milestone 3 – Year 3: Integrated Resilience and Pilot Demonstration** Conduct pilot-scale validation of the security solutions on realistic manufacturing setups

4. R&R (Roles and Responsibilities)

(max 10 pages)

Expected R&R of Research Institute	The CPSec Lab at Georgia Tech, led by Saman Zonouz, will be responsible for the design, development, and evaluation of the core cyber-physical security technologies proposed in this project. This includes the following tasks: ● Development of AI-based side-channel attack models and corresponding defense mechanisms for protecting manufacturing intellectual property, including real-time detection and mitigation strategies. ● Design and implementation of automated vulnerability assessment and criticality-aware analysis tools for industrial devices, controllers, and manufacturing networks. ● Development of cyber-physical impact assessment and automated recovery mechanisms, ● Integration of the proposed security components into a unified framework and validation through software prototypes, simulation, and experimental evaluation on representative manufacturing testbeds. The CPSec Lab has prior experience and ongoing research in cyber-physical system security, manufacturing security, industrial control systems, and side-channel analysis, with several research outcomes currently at TRL 3–4. These existing prototypes and validated concepts provide a strong basis for advancing the proposed technologies to higher TRLs through integration and experimental evaluation. If successful, the funds from this research will support graduate students working on manufacturing security and resilience, including opportunities for collaboration.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for: - Serve in an Industry Advisory Board (IAB)-like role to guide the project toward practical and deployable solutions. - Provide feedback on threat models, system assumptions, and technical design choices based on real manufacturing operations. - Validate the realism and relevance of attack scenarios, security objectives, and recovery strategies. - Share information on deployment challenges, including integration with existing equipment, performance constraints, reliability requirements, and operational workflows. - Advise on approaches to overcome deployment barriers, such as configuration practices, phased adoption strategies, and operator training needs.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes will include: 1. Development of an integrated cyber-physical security framework that protects manufacturing intellectual property, identifies critical workflows, and enables automated recovery in Industry 4.0 environments. 2. Advancement of existing TRL 3–4 research prototypes to higher TRLs through system

	integration and validation on representative manufacturing platforms. 3. Reduced operational risk by minimizing downtime, preventing product quality degradation, limiting equipment damage, and improving safety during cyber incidents. 4. Improved situational awareness and faster response for manufacturing operators through automated impact assessment and guided recovery actions. 5. Practical, deployable security solutions for Korean manufacturing SMEs, supporting secure adoption of smart manufacturing technologies and enabling future technology transfer and commercialization.

6. References

(1) Twisha Chattopadhyay, Fabricio Ceschin, Marco E. Garza, Dymytriy Zyunkin, Animesh Chhotaray, Aaron P. Stebner, **Saman Zonouz**, and Raheem Beyah, "One Video to Steal Them All: 3D-Printing IP Theft through Optical Side-Channels," *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, 2025.

(2) Sizhuang Liang, Xirui Peng, H. Jerry Qi, **Saman Zonouz**, and Raheem Beyah, "A Practical Side-Channel Based Intrusion Detection System for Additive Manufacturing Systems," *Proceedings of the 2021 IEEE 41st International Conference on Distributed Computing Systems (ICDCS)*, pp. 1075–1087, IEEE, 2021.

(3) Sriharsha Etigowni, Shamina Hossain-McKenzie, Maryam Kazerooni, Katherine Davis, and **Saman Zonouz**, "Crystal (Ball): I Look at Physics and Predict Control Flow! Just-Ahead-of-Time Controller Recovery," *Proceedings of the 34th Annual Computer Security Applications Conference (ACSAC)*, 2018.

(4) Aolin Ding, Matthew Chan, Amin Hass, Nils Ole Tippenhauer, Shiqing Ma, and **Saman Zonouz**, "Get Your Cyber-Physical Tests Done! Data-Driven Vulnerability Assessment of Robotic Aerial Vehicles," *Proceedings of the 2023 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, pp. 67–80, IEEE, 2023.

(5) Pengfei Sun, Luis Garcia, Gabriel Salles-Loustau, and Saman Zonouz, "Hybrid Firmware Analysis for Known Mobile and IoT Security Vulnerabilities," *Proceedings of the 2020 50th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, pp. 373–384, IEEE, 2020.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents

(max 10 pages)

	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization